Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **production AI systems**, **Python/C++ services**, distributed infrastructure, and reliable performance-focused APIs.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Advanced NLP, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant FastAPI inference workflows with asynchronous workers, explicit job states, failure isolation, and monitoring, processing 1K+ AI jobs at 3x throughput across **reliable distributed production services**.
- Architected a multi-tenant Azure SQL and cloud-storage platform for 500+ users, reducing dashboard latency 71% through **indexed queries, workload profiling, data isolation, and performance-focused backend optimization in production**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask features for uploads, model selection, grading history, and feedback, supporting 10K+ daily requests through **production REST APIs, customer-focused iteration, and end-to-end feature ownership**.
- Engineered concurrent Python ingestion with ThreadPoolExecutor, OCR, timeouts, and multimodal model fallbacks, processing 1K+ files per batch while reducing turnaround 60% through **testing, profiling, and fault isolation**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch Decision Transformer for offline reinforcement learning, processing 100K+ trajectory logs through reproducible pipelines and improving planning 18% through **model training, evaluation, and performance experimentation**.
- Integrated HD-map geometry and ego trajectories through spatial-temporal encoding and multi-head attention, improving closed-loop stability 15% while **debugging model, simulation, and control failures across reproducible autonomy experiments**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized distributed APIs and data pipelines, improving response times 30% through object-oriented patterns, AWS Glue, S3, caching, and **systematic profiling across cloud and cross-platform execution environments at scale**.
- Automated builds and 20+ unit and integration suites across Windows, Linux, and macOS using Jenkins, PyTest, Mockito, and virtual machines, strengthening **CI/CD, regression coverage, and release reliability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, React interfaces, and scalable real-time distributed workflows across production locations**.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while applying unit tests, Sentry, Azure Monitor, and Kibana for **observability, troubleshooting, and maintainable production operations across services**.

Skills

Core: Python, C++, TypeScript, JavaScript, Java, SQL, Data Structures, Algorithms, Operating Systems, OOP
AI/ML: PyTorch, Transformers, LLMs, Inference, Model Training, Fine-Tuning, Reinforcement Learning, RAG, Embeddings, FAISS, Evaluation
Platform: FastAPI, Flask, REST APIs, PostgreSQL, Redis, AWS, Azure, GCP, Kubernetes, Docker, Linux, Distributed Systems
Engineering: CI/CD, Git, Jenkins, Profiling, Observability, Monitoring, Logging, Unit Testing, Integration Testing, Debugging, Fault Isolation, API Design

Projects

InferenceForge | *Python, C++, PyTorch, FastAPI, Kubernetes, Redis, Prometheus*

- Designed a **distributed model-serving platform** with dynamic batching, request scheduling, KV-cache management, autoscaling, health checks, and Prometheus tracing for reliable, observable, performance-focused inference across Kubernetes workers under variable workloads.

EvalLoop | *Python, TypeScript, React, RAG, LLMs, PostgreSQL, Docker*

- Built an **end-to-end agentic evaluation application** with retrieval, structured tool calls, trace capture, regression datasets, automated scoring, failure analysis, and a React dashboard for comparing model behavior.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **semantic retrieval and RAG platform** over 50K+ documents using LLaMA-2, FAISS, and Pinecone, enabling contextual question answering and summarization with sub-second retrieval through optimized embedding pipelines.